



<Cipher/>

DevOps Roadmap

A structured, beginner-friendly path from fundamentals to production systems

13 STAGES · PREREQUISITES → ADVANCED · SELF-PACED



Roadmap Overview

PREREQUISITES → FUNDAMENTALS → CORE DEVOPS TECHNOLOGIES → ADVANCED

01

Prerequisites

02

Linux Fundamentals

03

Networking

04

Git & GitHub

05

Scripting & Automation

06

Containers

07

CI/CD

08

Cloud Fundamentals

09

Monitoring & Observability

10

Kubernetes

11

Infrastructure as Code

12

Configuration Mgmt

13 **ADVANCED**

DevSecOps · SRE · Platform Engineering · Cloud Architecture

Prerequisites

Optional groundwork before starting the roadmap. A basic grasp of programming logic and how backend systems work makes everything else easier to follow.

WHAT YOU'LL LEARN

- Programming Basics
- Backend Overview (optional)

CONCEPT FLOW



1. Linux Fundamentals

Linux runs most servers and DevOps tooling. Comfort with the command line is the foundation everything else in this roadmap builds on.

WHAT YOU'LL LEARN

- Linux Basics
- Files & Directories
- Users & Permissions
- Processes & Services
- Package Management
- Networking Commands
- Shell Basics



ESTIMATED TIME

2–3 Weeks

CONCEPT FLOW

Terminal



Shell



Processes



Filesystem

YOUTUBE-COURSE

[liunix redhat admin 1 - Abeer Hosni](#)

2. Networking Fundamentals

DevOps engineers constantly reason about how systems talk to each other. Solid networking knowledge is essential for debugging and designing infrastructure.

WHAT YOU'LL LEARN

- OSI Model
- DHCP
- TCP/IP
- HTTP / HTTPS
- IP Addressing
- Ports & Firewalls
- Subnetting
- DNS



ESTIMATED TIME

2-3 Weeks

CONCEPT FLOW

Client

DNS

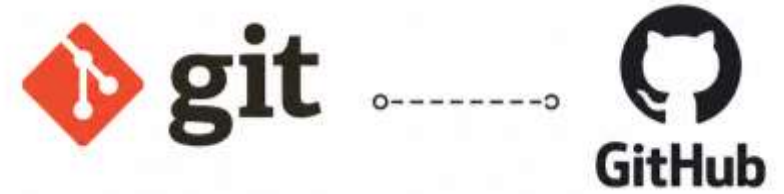
Network

Server

YOUTUBE-COURSE

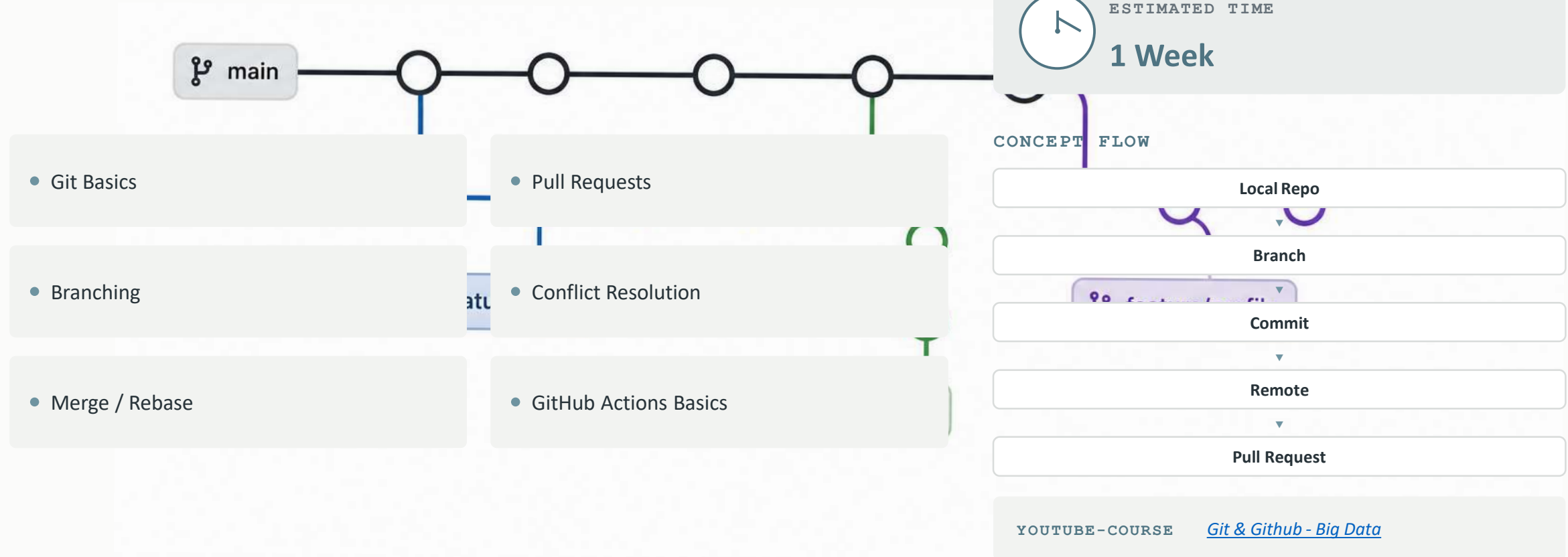
[IT, Cloud, DevOps and Network](#)

3. Git & GitHub



Version control is how teams collaborate on code and infrastructure safely. Git and GitHub are used daily in almost every DevOps workflow.

WHAT YOU'LL LEARN





4. Scripting & Automation

Automation is at the heart of DevOps. Shell scripting removes repetitive manual work and lets you build reliable operational tooling.

WHAT YOU'LL LEARN

- Bash

- Automation Scripts

- Environment Variables

- Cron Jobs



ESTIMATED TIME

1–2 Weeks

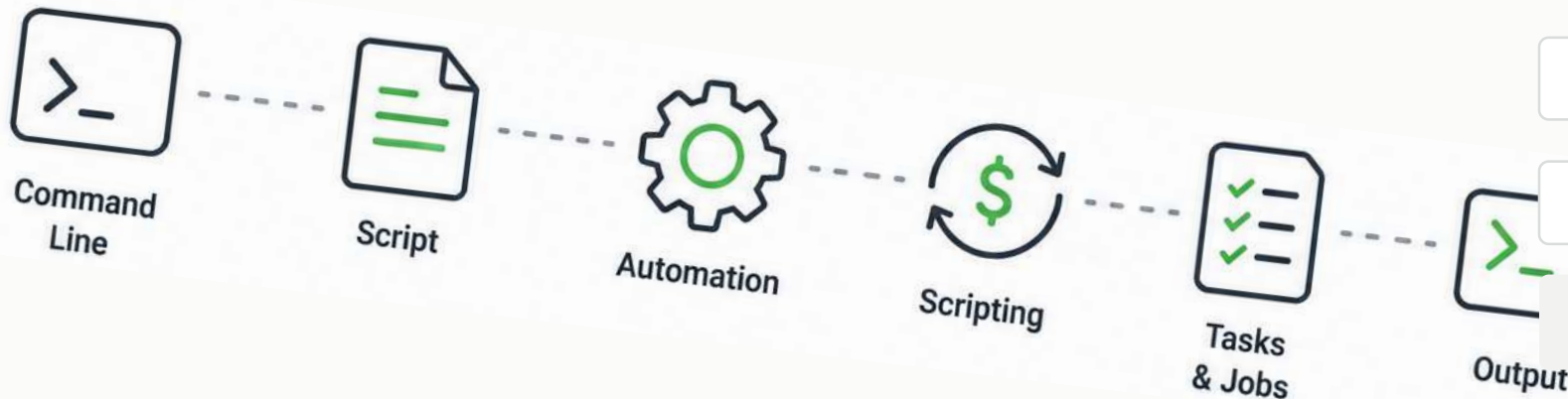
CONCEPT FLOW

Task

Bash Script

Cron Schedule

Automated Run



YOUTUBE-COURSE

[Bash \(crash course\)](#)

5. Containers

Containers package applications with everything they need to run consistently. Docker is the standard tool for building and running them.

WHAT YOU'LL LEARN

- Docker Basics

- Images

- Containers

- Dockerfile

- Volumes

- Networks

- Docker Compose

- Container Registries



ESTIMATED TIME

2–3 Weeks

CONCEPT FLOW

Application**Image****Container**

YOUTUBE-COURSE

[Docker - Big Data](#)

6. CI/CD

Continuous integration and delivery automate how code moves from a commit to a running system, catching problems early and shipping faster.

WHAT YOU'LL LEARN

- CI/CD Concepts

- Pipelines

- Build

- Test

- Artifact Management

- Deployment Automation

- GitHub Actions

- Jenkins / GitLab CI Basics



CI/CD
with
GitHub Actions



ESTIMATED TIME

2 Weeks

CONCEPT FLOW

Code

Build

Test

Artifact

Deploy

YOUTUBE-COURSE

[Githu Actions \(ci/cd\) - ELfakharany](#)

7. Cloud Fundamentals

Modern infrastructure largely lives in the cloud. Understanding core cloud services helps you design, deploy, and scale systems reliably.

WHAT YOU'LL LEARN

- Cloud Concepts

- Compute

- Storage

- Networking

- IAM

- Databases

- AWS



ESTIMATED TIME

4–5 Weeks (SAA)

CONCEPT FLOW

Users



Compute



Storage & Network



IAM

YOUTUBE-COURSE

[AWS SAA - AWS Riyadh User Group](#)

8. Monitoring & Observability

You can't fix what you can't see. Monitoring and observability tools give visibility into system health, performance, and failures.

WHAT YOU'LL LEARN

- Monitoring Concepts

- Metrics

- Logs

- Alerts

- Prometheus

- Grafana

- Loki



ESTIMATED TIME

1–2 Weeks

CONCEPT FLOW

Application**Metrics & Logs****Prometheus / Loki****Grafana**

SEARCH

Search: Grafana, Loki, Prometheus, exporters...

9. Kubernetes

Kubernetes orchestrates containers at scale, handling deployment, scaling, and recovery automatically across a cluster of machines.

WHAT YOU'LL LEARN

- Kubernetes Concepts

- Pods

- Deployments

- Services

- ConfigMaps / Secrets

- Ingress

- Volumes

- Namespaces

- Helm Basics



ESTIMATED TIME

3–4 Weeks

CONCEPT FLOW

Users



Ingress



Service



Pods



Cluster

YOUTUBE-COURSE

[Kubernetes - Codographia](#)

10. Infrastructure as Code

IaC lets you define and provision infrastructure through code instead of manual steps, making environments repeatable and version-controlled.

```
resource "aws_instance" "web" {  
  ami = "ami-0abc1234"  
  instance_type = "t3.micro"  
}
```

- IaC Concepts

- Variables

- Terraform

- State

- Providers

- Modules

- Resources



ESTIMATED TIME

2 Weeks

CONCEPT FLOW

Code



Terraform



Infrastructure

YOUTUBE-COURSE

[TERRAFORM - DolphinED](#)

STAGE 12

11. Configuration Management

```
hosts: all
tasks:
  - name: install
    apt:
      name: nginx
```

YAML

WHAT YOU'LL LEARN

- Configuration Mgmt Concepts

- Ansible

- Inventory

- Playbooks

- Roles

Configuration management tools keep servers and environments consistent by automating setup and ongoing configuration at scale.



ESTIMATED TIME

1-2 Weeks

CONCEPT FLOW

Playbook

Inventory

Servers

YOUTUBE-COURSE

[Ansible - kodeclod](#)

STAGE 12

Advanced

Once the core roadmap is complete, these areas are where DevOps careers typically specialize next. Each is a deep discipline on its own, worth exploring once the fundamentals feel solid.

01

DevSecOps

Security built into every stage of the pipeline.

02

SRE

Reliability, uptime, and incident response at scale.

03

Platform Engineering

Internal tooling that other teams build on.

04

Cloud Architecture

Designing scalable, resilient cloud systems.